Naveen Hemanth Kokkonda

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Education

Sreyas Institute of Engineering and Technology — Hyderabad, India
B.Tech, Computer Science and Engineering (Data Science) - CGPA: 7.51/10 — *Oct. 2022 – Jun. 2026*

Experience

AI Engineer (Full Stack) — Dec. 2025 – Present
Tradl AI - Stock Research and Trading Platform — *On-site, Bangalore*

- Built and shipped the Home Page and MarketBuzz News Page end-to-end, the platform's two primary user-facing surfaces, from design to production
- Cut initial render time 40% and heap usage 60% on financial data tables by virtualizing 1,000+ live stock tickers with TanStack Virtual, extended to the Discovery and Playground tables
- Deployed the Pattern Detection frontend on AWS with a live WebSocket feed streaming real-time ticker bars and LTP (last traded price) directly onto TradingView charts
- Built a citation-grounded financial RAG system for question answering over PDF filings using Python, Chainlit, Qdrant, and DuckDB
- Engineered hybrid retrieval combining Qwen3-Embedding-4B dense vectors with sparse search and Qwen3-Reranker-4B cross-encoder reranking
- Orchestrated an agentic workflow over document retrieval, structured financial-metric lookup, and multi-step reasoning, routing across Claude Sonnet/Haiku and Bedrock Qwen3-235B by task complexity
- Designed ingestion and indexing pipelines with Qwen3-VL page-level extraction for charts and tables, processing 45 financial filings into 30,137 indexed points and lifting retrieval recall from 0.37 to 0.61 on a 90-question financial QA benchmark

Projects

TOON Studio - Token Efficiency Toolkit | Live Demo | GitHub — Dec. 2025

Technologies: Python, FastAPI, React, WebSockets, Google Gemini AI, LangSmith, Vite

- Created a full-stack JSON/TOON conversion tool to cut LLM API token usage, with a transparent cost analytics dashboard showing per-token pricing ($0.15/1M input, $0.60/1M output).
- Integrated Google Gemini 2.5 Flash with LangSmith tracing and WebSocket streaming for real-time, AI-powered data analysis with live token usage monitoring.
- Optimized frontend performance with debounced API calls and a Notion-inspired UI, deployed on Vercel with a CI/CD pipeline.

KnoRa AI Knowledge Assistant | Live Demo | GitHub — Aug. 2025

Technologies: Python, Streamlit, FAISS, Sentence Transformers, Groq API, Docker, RAG

- Engineered a RAG system enabling semantic document search across multiple formats (PDF, DOCX, XLSX, PPTX, CSV, MD, TXT) with intelligent chunking and context overlap.
- Implemented a FAISS vector store using all-MiniLM-L6-v2 embeddings with batch processing and configurable similarity thresholds for efficient retrieval.
- Integrated the Groq LLM API with multiple model options, featuring streaming responses, temperature control, and source citation.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL
Web: React, FastAPI, Streamlit, REST APIs, WebSockets, TanStack Query/Virtual
AI & ML: LLM APIs (Claude, GPT, Gemini, Qwen, LLaMA), RAG Systems, Agentic Workflows, Hybrid Retrieval, Reranking, LangChain, LangSmith, Langfuse, FAISS, Qdrant
Data & Cloud: DuckDB, MongoDB, AWS (EC2, S3, Bedrock), Vercel
Tools: Git, Docker, CI/CD, Postman

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **(Aug 2025)** [View Certificate]